

Parsons Software Developer Intern - What to Learn

Why these keywords

This JD is a generalist full-stack internship: it names a Microsoft-centric stack (.NET, C#, SQL Server) alongside open-web basics (Node.js, HTML/CSS/JS) and lists Angular as an alternative framework to React. Your React/JS background transfers directly to most of it – the delta is mainly syntax and platform conventions, not new concepts.

MUST learn (Class A – explicitly required/listed)

.NET & C# – Microsoft’s back-end framework/language, the ASP.NET Core flavor is the relevant one for web APIs. Bridge: it’s OOP like Java/C++, and ASP.NET Core Web API controllers map almost 1:1 to Express.js route handlers you already write (routing, middleware, dependency injection instead of manual imports). 20% to know: create a minimal Web API project (‘dotnet new webapi’), write a GET/POST controller, understand ‘appsettings.json’ config and Entity Framework Core as the ORM (like Prisma/Mongoose, but for SQL Server).

SQL Server – Microsoft’s relational database. You already know SQL (Skimly project) and Postgres/MongoDB – the query language (T-SQL) is 95% standard SQL with a few syntax differences (‘TOP’ instead of ‘LIMIT’, ‘GETDATE()’ instead of ‘NOW()’). 20%: install SQL Server Express or use Azure Data Studio, run basic CRUD + JOIN queries, know what a stored procedure is.

Angular – a full front-end framework (like React but more opinionated/batteries-included: built-in DI, RxJS observables, two-way binding). Bridge: components + templates + services map to your React components + hooks + custom hooks/context. 20%: build one component with ‘@Component’, understand ‘*ngFor’/‘*ngIf’ directives (JSX map/ternary equivalents), and services with ‘@Injectable’ (like a custom hook that fetches data).

Java – If asked, it’s one of several acceptable ”server-side languages” (C#, Java, Node, Python) – you don’t need deep Java, just enough to say you can read/write it. 20%: syntax basics (classes, interfaces, generics) – if you know C++ and JS, Java is very approachable.

Microsoft Power Platform – low-code tool suite (Power Apps, Power Automate, Power BI) for building internal business apps and workflow automation without full custom code. 20%: know it exists as a ”citizen developer” tool for automating workflows/forms – unlikely to be asked for deep expertise as an intern, but be ready to say ”no hands-on experience yet, but conceptually similar to low-code/no-code automation.”

GOOD to know (Class B – process/practices, already partly covered)

Version control – you already use Git daily; just be ready to talk about branching/PR workflows if asked.

Security features (web) – basic web security hygiene: input validation, HTTPS, auth (JWT/OAuth – you already know this), avoiding SQL injection (parameterized queries – you already do this via Prisma/raw parameterized Neon SQL in Skimly).

Data problem-solving – your 1000+ DSA / CodeChef background directly covers this; no new prep needed.

Priority if time is short

1. .NET/C# basics (biggest gap, most JD emphasis). 2. SQL Server / T-SQL syntax differences from Postgres/MySQL. 3. Angular fundamentals (only if interview mentions it specifically). 4. Power Platform – just enough to discuss conceptually.